Video Reasoning through the Video-of-Thought (VoT) Framework

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Motivation and Scope

Challenges Driving Advanced Video Reasoning

- Long-term dependencies and occlusions hinder comprehension.
- Fine-grained spatial-temporal grounding remains unreliable.
- Causal and social queries exceed current cognitive modeling.
  - Video MLLMs advance, yet underuse Chain-of-Thought in video.
  - MotionEpic targets complex perception-to-reasoning gaps.
- Combines STSG grounding with a VoT stepwise reasoning path.

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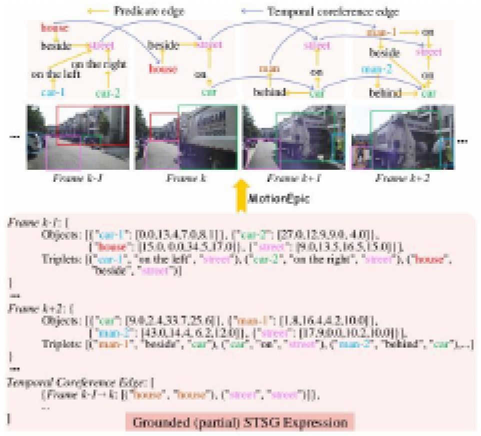
Related Work Landscape

Related Work: STSG for Video Reasoning



- STSG models objects, predicates, temporal coreference
  - Supports compositional multi-step video reasoning
  - Captures object relations across successive frames
- Bridges spatial layout with temporal dynamics
  - Exposes MLLM limits: weak visual grounding
  - Temporal logic often shallow in prior systems

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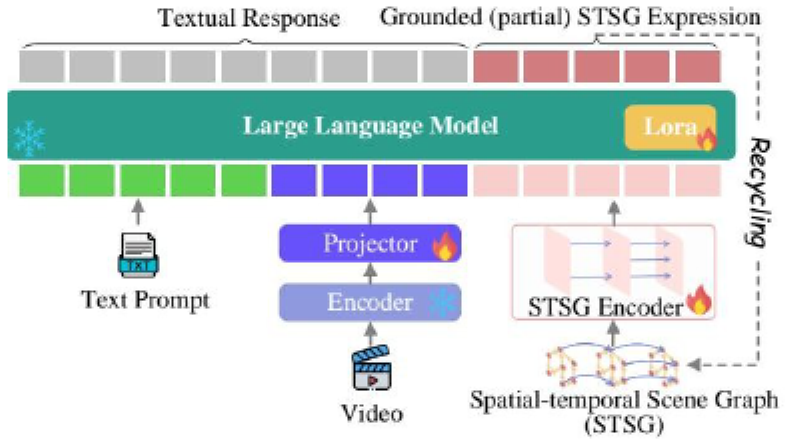
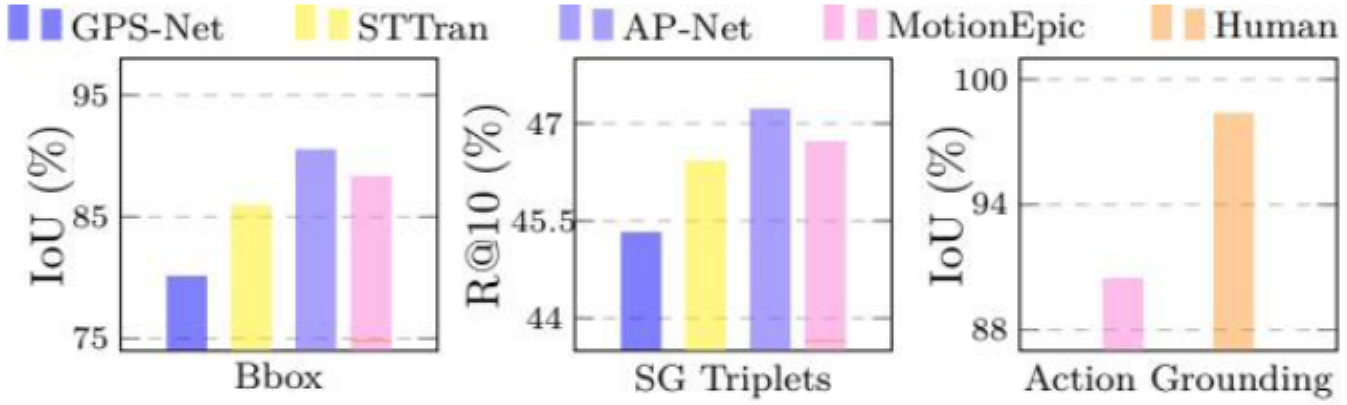
MotionEpic: STSG-grounded MLLM

MotionEpic Architecture Overview



- Vicuna-7B backbone fuses prompt, video, and STSG.
  - ViT-L/14 + Q-Former encode frames for perception.
  - Projectors align video/text embeddings for fusion.
- STSGs refined to lighten compute and aid reasoning.
  - STSGs built and fused via a Recurrent Graph Transformer.
  - Grounded expressions recycled to refine reps, responses.

MotionEpic: Precise Grounding



- Focuses on precise spatial-temporal grounding
  - Outperforms GPS-Net, STTran, AP-Net with grounding-aware tuning
  - Optimizes both coarse- and fine-grained correspondences
- No need for external STSG annotations or supervision

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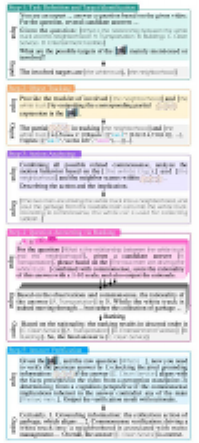
Video-of-Thought (VoT) Framework

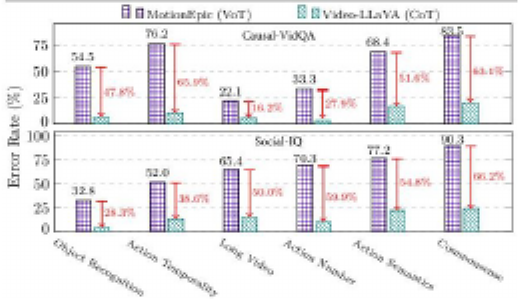
10 VoT: Targets to Verified Answers



- Define the task and extract target entities
  - Track objects across frames with identities
  - Analyze actions and interactions over time
- Score and rank candidate answers by likelihood
  - Verify answers using grounded visual evidence
  - Ensure robust perception-to-cognition flow

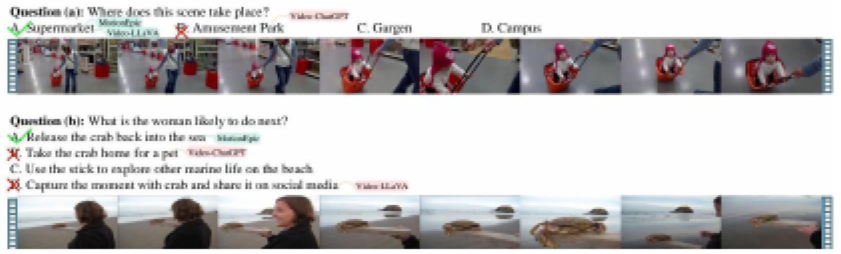
0 Experiments: VoT vs CoT vs Human

Data	CoT		VoT	Human
	Video-LLaVA	MotionEpic	MotionEpic	
Causal-VidQA	32.4	56.8	74.3	80.6
Social-IQ	22.3	40.1	61.4	72.7



- VoT sets SOTA; lower error than CoT on Causal-VidQA, Social-IQ.
  - Approaches human performance across challenging categories.
  - Zero-shot: robust generalization; clear gains over CoT.
- Verification mechanisms are crucial to sustaining accuracy.

12 Grounded Multi-step Video QA Examples



- Frames, options, and rationales visualized
  - Targets identified and tracked across shots
  - Actions analyzed with explicit temporal links
- Commonsense cues explain correct vs. distractors
  - Candidate answers scored using grounded evidence
  - Verified choices highlight stepwise VoT reasoning

13 Conclusion and Future Directions

- STSG-grounded MotionEpic advances video reasoning.
- VoT structures multi-step QA for reliability and trust.
- Consistent gains over CoT and strong prior baselines.
  - Integrates spatial-temporal modeling with reasoning.
  - Future: richer commonsense priors; longer videos.
- Few/zero-shot adaptation and broader benchmarks.